WORKED SOLUTION

STEP III 1995, Question 9

Mechanics • Compound Pendulum

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Read the question before you go on. It is in the STEP Support Programme's question database:

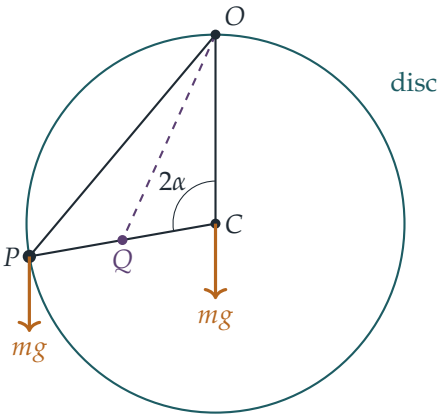
step.maths.org/questions/95-s3-q9

The key idea

The disc and the particle have the *same* mass, so the centre of mass of the whole system is the midpoint Q of CP. Almost everything follows from that: the stable equilibrium has Q directly below O, and the energy equation in part (iii) only needs the height of Q. The rest is the geometry of the isosceles triangle OCP and a standard moment of inertia integral.

Solution

We model the disc as a rigid lamina and P as a particle fixed to its rim; the axis is smooth, so mechanical energy is conserved. Throughout, $OC = CP = r$ and $\angle OCP = 2\alpha$.



Part (i)

Since the disc (centre of mass C) and the particle (at P) both have mass m, the centre of mass of the system is the midpoint Q of CP. The only forces are gravity and the reaction at the axis, so the system is in equilibrium exactly when Q is vertically in line

with O, and the equilibrium is stable when Q is *below* O (any displacement then raises Q, increasing the potential energy). So in stable equilibrium OQ is vertical, and β is the angle $\angle COQ$.

Use axes in the plane of the disc with origin C and O at $(r, 0)$; then $P = r(\cos 2\alpha, \sin 2\alpha)$ and $Q = \frac{r}{2}(\cos 2\alpha, \sin 2\alpha)$. Hence

$$\overrightarrow{OC} = (-r,\, 0), \qquad \overrightarrow{OQ} = \tfrac{r}{2}(-(2 - \cos 2\alpha),\ \sin 2\alpha).$$

Since $2 - \cos 2\alpha > 0$, the vector $\overrightarrow{OQ}$ has a positive component $\frac{r}{2}(2 - \cos 2\alpha)$ along $\overrightarrow{OC}$ and a component $\frac{r}{2}\sin 2\alpha$ perpendicular to it, so β is acute and

$$\tan\beta = \frac{\frac{r}{2}\sin 2\alpha}{\frac{r}{2}(2 - \cos 2\alpha)} = \frac{\sin 2\alpha}{2 - \cos 2\alpha},$$

as required.

Part (ii)

We take $\rho x / r$ to be mass per unit area. The ring between radii x and $x + \mathrm{d}x$ has area $2\pi x\,\mathrm{d}x$, so

$$m = \int_0^r \frac{\rho x}{r}\, 2\pi x\,\mathrm{d}x = \frac{2\pi\rho r^2}{3}.$$

Every point of that ring is distance x from the axis through C perpendicular to the disc, so the moment of inertia about that axis is

$$I_C = \int_0^r \frac{\rho x}{r}\, 2\pi x \cdot x^2\,\mathrm{d}x = \frac{2\pi\rho r^4}{5} = \frac{3}{5}mr^2,$$

using $2\pi\rho r^2 = 3m$. The axis through O is parallel to this one (both are perpendicular to the vertical plane of the disc) at distance $OC = r$, so by the parallel axis theorem

$$I_O = \frac{3}{5}mr^2 + mr^2 = \frac{8}{5}mr^2.$$

Part (iii)

Moment of inertia of the system. Triangle OCP is isosceles with apex angle 2α, so the chord is $OP = 2r\sin\alpha$. The particle adds $m(2r\sin\alpha)^2$, giving

$$I = \frac{8}{5}mr^2 + 4mr^2\sin^2\alpha = \frac{4}{5}mr^2\,(2 + 5\sin^2\alpha).$$

Drop of the centre of mass. At release OQ is horizontal, so Q is level with O. In the final position P is vertically below O, so OP is vertical, and the depth of any point below O is its distance along OP from O.

- P is at depth $OP = 2r\sin\alpha$.
- The base angles of triangle OCP are $\frac{1}{2}(\pi - 2\alpha) = \frac{\pi}{2} - \alpha$, so $\angle COP = \frac{\pi}{2} - \alpha$ and C is at depth $r\cos(\frac{\pi}{2} - \alpha) = r\sin\alpha$.

Q is the midpoint of CP, so its depth is the average, $\frac{3}{2}r\sin\alpha$. The system (total mass $2m$) has therefore lost potential energy $2mg\cdot\frac{3}{2}r\sin\alpha = 3mgr\sin\alpha$.

Ask yourself Does the system actually reach the position with P below O?

Released from rest with Q level with O, the system swings like a compound pendulum between the two positions with OQ horizontal, passing through every position in which Q is below the level of O. In the final position Q is at depth $\frac{3}{2}r\sin\alpha > 0$, so it is reached (and the condition that C starts lower than P just fixes which way the swing goes).

Energy. Starting from rest, with angular speed ω,

$$\tfrac{1}{2}I\omega^2 = 3mgr\sin\alpha \quad\Longrightarrow\quad \omega^2 = \frac{6gr\sin\alpha}{\frac{4}{5}r^2(2+5\sin^2\alpha)} = \frac{15g\sin\alpha}{2r(2+5\sin^2\alpha)}.$$

C moves on a circle of radius $OC = r$ about O, so $v = r\omega$ and

$$v^2 = \frac{15gr\sin\alpha}{2(2+5\sin^2\alpha)},$$

as required.

Maximum. Put $s = \sin\alpha$; since $0 < \alpha \le \pi/2$, $s \in (0,1]$. Let $f(s) = \dfrac{s}{2+5s^2}$. Then

$$f'(s) = \frac{(2+5s^2) - 10s^2}{(2+5s^2)^2} = \frac{2-5s^2}{(2+5s^2)^2},$$

which is positive for $s < \sqrt{2/5}$ and negative for $s > \sqrt{2/5}$. As $\sqrt{2/5} < 1$ lies in the allowed range, f has its maximum there, $f = \dfrac{\sqrt{2/5}}{4} = \dfrac{\sqrt{10}}{20}$. Hence

$$v^2_{\max} = \frac{15gr}{2}\cdot\frac{\sqrt{10}}{20} = \frac{3\sqrt{10}}{8}\,gr \approx 1.19\,gr, \qquad \text{when } \sin\alpha = \sqrt{2/5}.$$

Checks

- If $\alpha \to 0$, P sits at O and the disc alone hangs with OC vertical: indeed $\tan\beta \to 0$. If $\alpha = \pi/2$, P is diametrically opposite O, Q lies on OC and again $\beta = 0$.
- With $\alpha = \pi/2$: $I = \frac{28}{5}mr^2$, drop $\frac{3}{2}r$ for mass $2m$, so $v^2 = r^2\cdot 3mgr\cdot\frac{10}{28mr^2} = \frac{15}{14}gr$, matching the formula.
- We integrated the equation of motion numerically (RK4) from the release position for several α; for example at $\alpha = 0.7$, $r = 1.3$, $g = 9.81$ the simulation gives $v^2 = 15.1206$, agreeing with the formula to six figures. A direct search also confirms the maximum $3\sqrt{10}/8 \approx 1.1859$ (in units of gr).

Common slips

- Using the uniform-disc value $\frac{1}{2}mr^2$ for I_C; the density here increases outwards, which pushes I_C up to $\frac{3}{5}mr^2$.

- Forgetting the particle's contribution $m \cdot OP^2$ to the moment of inertia in part (iii).
- Giving ω^2 instead of v^2: the question asks for the speed of C, which is $r\omega$.

Beyond STEP

A rigid body pivoted about a horizontal axis is a *compound pendulum*: it behaves like a simple pendulum whose bob is at the centre of mass, but with inertia I instead of Md^2. Small oscillations about the equilibrium in part (i) have period $2\pi\sqrt{I/(Mgd)}$, where $d = OQ$, and the "equivalent simple pendulum" length $I/(Md)$ leads to Kater's reversible pendulum, once used to measure g precisely. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and the parallel axis theorem in this spirit.

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